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Quantifying the Consistency of Neural Signals: A Subject-Prevalence Analysis of Neuroimaging Data

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Abstract

Functional magnetic resonance imaging group analyses are commonly used to detect population-level effects, yet they do not directly quantify how consistently a signal is expressed across individuals. This thesis develops a statistical framework for voxel-wise subject-prevalence inference, targeting a complementary question: at each voxel, how many subjects show evidence of activation. The proposed approach formalizes prevalence through a u-out-of-N perspective and translates it into a practical inferential procedure suitable for high-dimensional neuroimaging data.

Valid inference is achieved via a closed-testing construction within an exploratory multiple-testing paradigm, enabling post-hoc, data-driven exploration while retaining a global confidence guarantee. The main output is a 3D map of simultaneous lower confidence bounds on prevalence, interpretable as certified minimum numbers of active subjects at each voxel. This representation supports transparent reporting across scientific thresholds u, clarifying how conclusions change from subgroup-level evidence to effects shared by a majority of participants.

The methodology is illustrated using publicly available subject-level statistical maps from NeuroVault collection #2447 for a noun/verb judgement contrast. The results show that as the prevalence threshold increases, the detected brain regions shrink, highlighting the difference between effects seen in most subjects and effects limited to smaller groups. Overall, voxel-wise subject-prevalence mapping provides an interpretable and statistically principled complement to standard group-level inference for studying neural-signal consistency in the presence of inter-subject heterogeneity.

Keywords: closed testing; functional magnetic resonance imaging; multiple testing; neuroimaging; subject prevalence.

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List of Abbreviations

ARI All-Resolutions Inference

BOLD Blood Oxygenation Level-Dependent

CDF Cumulative Distribution Function

FDP False Discovery Proportion

FDR False Discovery Rate

fMRI functional Magnetic Resonance Imaging

FWER Familywise Error Rate

GLM General Linear Model

NIfTI Neuroimaging Informatics Technology Initiative

ROI Region of Interest

TDP True Discovery Proportion

Chapter 1

Introduction

Neuroimaging serves as the primary discipline for the in vivo depiction of the anatomy and function of the central nervous system (Fulham, 2004). With the steady increase in neuroimaging data availability and computational power, it is now possible to investigate neural behavior at progressively higher spatial resolutions (Pinho et al., 2018). By linking observed Blood Oxygenation Level-Dependent (BOLD) signals to underlying cognitive processes, researchers can map regional specialization and functional connectivity with unprecedented detail (Lindquist, 2008). The basic intuition is that neural activity increases local metabolic demand, which in turn elicits greater delivery of oxygenated blood to the engaged region, so that relative increases in oxygenation are taken as an indirect index of stimulus-evoked activation. However, as the field moves toward large-scale group studies and clinical applications, a significant challenge remains: ensuring that detected signals are not only statistically significant, but also consistent across individuals. This thesis addresses the gap between individual-level observations and population-level statistical inference. By quantifying signal prevalence, which means quantifying the number of subjects in a study that have a true signal, this research seeks to improve the reliability of critical network nodes and functional signatures in cognitive neuroscience.

1.1 Overview of neuroimaging

The fundamental building block of brain mapping with functional Magnetic Resonance Imaging (fMRI) is the localization of evoked brain responses to cognitive stimuli. Localization is typically performed by correlating a sequence of experimental tasks with measured BOLD signals in each brain region, testing statistical significance against an "inactive" null hypothesis (Lindquist, 2008). In a classical task-based fMRI study, we observe N subjects, each exposed to a sequence of stimuli over time. Brain activity is recorded through the BOLD signal, measured on a 3D grid of m volume units called voxels, where typically m is in the tens of thousands (Lindquist, 2008). In a group analysis, the aim is merging information from all subjects to test, for each voxel i, the null hypothesis H_i^{group} that the voxel is not active under the stimulus. From suitable pre-processing and first-level analysis, standard methods immediately provide overall t-statistics t_i^{group} and p-values p_i^{group}, that can then be arranged into single 3D maps. This methodology has evolved from simple voxel-wise testing to more complex cluster-based methods, which are currently the most common types of inference in the field (Rosenblatt et al., 2018).

1.2 Multiple testing and error rate control

Testing many thousands of voxels simultaneously introduces a severe multiplicity problem, where the probability of making a Type I error (false discovery) increases with the number of tests (Goeman and Solari, 2011). This issue requires the adoption of error rate controls, most notably Familywise Error Rate (FWER) and False Discovery Rate (FDR). In general, consider a collection of null hypotheses $\mathcal{M} = \{H_1, \ldots, H_m\}$. Denote by $\mathcal{T} \subseteq \mathcal{M}$ the unknown subset of true null hypotheses, i.e., inactive voxels in the case of fMRI analysis. For any subset $\mathcal{S} \subseteq \mathcal{M}$, we identify as false discoveries the number of true null hypotheses contained in the subset

$$\text{FD}(\mathcal{S}) = |\mathcal{S} \cap \mathcal{T}|.$$

In fMRI, this measure corresponds to the number of inactive voxels within a region of the brain. From this quantity, we can immediately derive the number of true

discoveries (active voxels)

$$\mathrm{TD}(\mathcal{S}) = |\mathcal{S}| - \mathrm{FD}(\mathcal{S}),$$

as well as the False Discovery Proportion (FDP) and True Discovery Proportion (TDP)

$$\mathrm{FDP}(\mathcal{S}) = \frac{\mathrm{FD}(\mathcal{S})}{|\mathcal{S}| \vee 1}, \qquad \mathrm{TDP}(\mathcal{S}) = \frac{\mathrm{TD}(\mathcal{S})}{|\mathcal{S}| \vee 1},$$

where $\vee$ denotes the maximum.

FWER-based methods (Bonferroni, 1936; Holm, 1979; Hommel, 1988) are the archetypical approach for confirmatory analysis, controlling the probability of making any false rejection at a prespecified rate $\alpha \in (0, 1)$. Such methods provide a rejection set $\mathcal{R} \subseteq \mathcal{M}$ with the guarantee that

$$\mathbb{P}\left(\mathrm{FD}(\mathcal{R}) > 0\right) \leq \alpha.$$

While rigorous, they are often considered too strict for exploratory research, where some false leads are expected.

In contrast, FDR-based methods (Benjamini and Hochberg, 1995; Benjamini and Yekutieli, 2001) offer a milder alternative by controlling the expected proportion of falsely rejected hypotheses among the total rejected set. They return a rejection set $\mathcal{R}$ such that

$$\mathrm{FDR}(\mathcal{R}) \coloneqq \mathbb{E}\left[\mathrm{FDP}(\mathcal{R})\right] \leq \alpha.$$

Notice that this control holds only in expectation, while the realized false discovery proportion may exceed α in individual experiments. Moreover, FDR control applies to the full collection of tested hypotheses $\mathcal{M}$ (the entire brain), but does not provide guarantees for arbitrary subsets $\mathcal{S}$ (brain regions, clusters, etc.), i.e., $\mathrm{FDR}(\mathcal{R} \cap \mathcal{S})$ may significantly exceed the nominal level α.

In fMRI analysis, one of the most common approaches to inference is cluster-extent based thresholding (Friston et al., 1994). Given a pre-specified threshold τ, the method considers all supra-threshold clusters, i.e., regions of connected voxels all having test statistics $t_i^{\mathrm{group}} \geq \tau$. In this framework, the researcher tests a

global null hypothesis for each cluster with FWER control at the level of clusters; a significant result implies that at least one voxel within the cluster contains a signal. Consequently, the method does not provide information on the specific quantity or precise location of the signal within a detected region ("spatial specificity paradox") (Rosenblatt et al., 2018). Furthermore, because FWER control is tied to the cluster level, any follow-up inference or "sub-clustering" within these regions is statistically invalid and would lead to inflated error rates (Kriegeskorte et al., 2009).

To resolve these issues, recent advances suggest shifting toward the TDP. Unlike traditional methods where the procedure dictates the rejected set based on a fixed error criterion, TDP-based methods allow the user to freely select any subset of hypotheses post-hoc. They then return a simultaneous confidence statement on the number of correct rejections within that specific set. More in detail, these procedures return lower $(1-\alpha)$-confidence bounds $\underline{\mathrm{TD}}(\mathcal{S})$ such that

$$\mathbb{P}\left(\mathrm{TD}(\mathcal{S}) \geq \underline{\mathrm{TD}}(\mathcal{S}) \text{ for all } \mathcal{S} \subseteq \mathcal{M}\right) \geq 1-\alpha. \tag{1.1}$$

Analogous lower confidence bounds for the TDP can be immediately derived as

$$\underline{\mathrm{TDP}}(\mathcal{S}) = \frac{\underline{\mathrm{TD}}(\mathcal{S})}{|\mathcal{S}| \vee 1}.$$

Moreover, upper $(1-\alpha)$-confidence bounds for the number and proportion of false discoveries are

$$\overline{\mathrm{FD}}(\mathcal{S}) = |\mathcal{S}| - \underline{\mathrm{TD}}(\mathcal{S}), \qquad \overline{\mathrm{FDP}}(\mathcal{S}) = \frac{\overline{\mathrm{FD}}(\mathcal{S})}{|\mathcal{S}| \vee 1}.$$

This flexibility ensures that researchers can quantify the risk of following up on specific subsets without compromising statistical validity. In fMRI analysis, such an approach allows one to quantify the proportion of active voxels within any brain region, even selected post-hoc, as well as to carry on follow-up inference inside promising regions.

This line of work started with methods that give high-probability (exceedance) control of the FDP (Genovese and Wasserman, 2006) and was later developed into a general framework for exploratory, post-hoc multiple testing based on the

closed testing principle (Marcus et al., 1976) by Goeman and Solari (2011). In neuroimaging, these ideas were adapted in the ARI framework (Rosenblatt et al., 2018). Since then, several extensions and alternatives for post-hoc inference on the number or proportion of true discoveries have appeared, including permutation-based methods based on critical vectors of ordered p-values (Blanchard et al., 2020; Blain et al., 2022; Andreella et al., 2023) or on global sum tests (Vesely et al., 2023).

1.3 Problem statement: inter-subject variability in group studies

Group-level analysis detects and quantifies signal by pooling data across subjects to estimate a mean (population-level) effect, which increases sensitivity but can mask substantial inter-subject variability. A complementary research question is *consistency*: whether the effect is reliably expressed across individuals (e.g., present in most subjects) rather than driven by a small subgroup.

A possible approach is partial conjunction hypothesis testing (Benjamini and Heller, 2008), which can improve robustness to inter-subject variability by providing a voxel-wise lower bound on the number of activated subjects, usually with control of the FDR. However, it is not ideally suited to fully exploratory analyses. First, as noted in the previous section, the error guarantee is an expectation, and so the realized fraction of false positives in any single study can fluctuate substantially and may still end up higher than the nominal target. Second, the guarantee is designed for the entire brain rejection set taken as a whole; it does not automatically extend to smaller regions or clusters chosen after inspecting the results, so it offers limited protection for the types of post-hoc "drill-down" questions that commonly arise in exploratory fMRI work.

This work aims at overcoming these constraints. The proposed methodology will ensure that confidence statements remain valid for any user-selected region, allowing rigorous follow-up inference within clusters and a more robust characterization of consistency between subjects.

1.4 Research aim: quantifying voxel-wise subject prevalence

The primary objective of this thesis is to develop and apply a closed-testing-based statistical framework that shifts neuroimaging inference from simple group averaging to the quantification of inter-subject signal consistency. Group-level analysis detects signal by pooling information across all subjects, but this averaging can mask substantial inter-subject variability. The present work focuses on the closely related issue of consistency across subjects: is the effect reliably expressed across individuals (e.g., present in most subjects) rather than driven by a small subgroup?

Concretely, we aim to identify brain regions that show reliable activation across subjects, in the sense that at least a pre-specified number u out of N individuals display an effect at that voxel. This u-out-of-N perspective is conceptually related to partial conjunction testing (Benjamini and Heller, 2008), but we pursue it by adapting the exploratory multiple-testing framework of Goeman and Solari (2011), which provides valid post-hoc guarantees under data-driven selection. By leveraging simultaneous confidence statements, the proposed approach supports exploratory workflows while controlling the statistical risk of "double-dipping" (selecting regions after seeing the data and then over-interpreting them), allowing researchers to inspect and refine sub-clusters without invalidating inference.

Methodologically, we develop a computational pipeline that produces voxel-wise prevalence summaries, yielding a more transparent view of group-level reliability than standard aggregation alone. For each voxel, we compute a simultaneous lower confidence bound on the number of subjects exhibiting activation. The simultaneity of these bounds across all voxels ensures valid inference even when clusters or regions are selected and reported post-hoc. This procedure yields a three-dimensional brain map in which the spatial distribution of activation consistency (or its absence) can be directly visualized. The methodology is illustrated using using high-resolution statistical maps obtained from the NeuroVault repository (Gorgolewski et al., 2015). Furthermore, we quantify how varying the subject-prevalence threshold u changes the spatial extent and localization of functional maps. Finally, we compare prevalence-based consistency mapping to traditional group aggregation to illustrate how inter-subject variability and outliers can affect the apparent location and extent of signal.

Manuscript structure. Chapter 2 formalizes the fMRI analysis setting and reviews ARI. Chapter 3 introduces the proposed closed-testing prevalence framework and its computational implementation. Chapter 4 presents the NeuroVault-based experiments, and Chapter 5 discusses implications, limitations, and directions for future work. Finally, the appendix presents additional results for the same dataset as in Chapter 4, based on a different contrast.

Chapter 2

Literature Review: fMRI group analysis and ARI

This chapter reviews the standard analysis pipeline in fMRI, explains why localization becomes difficult in group studies, and summarizes ARI (Rosenblatt et al., 2018), a post-hoc inference framework that supports valid quantification of activation within arbitrary user-selected regions. The next chapter will connect the methodology underlying ARI to the voxel-wise subject-prevalence objective of this thesis.

2.1 Structure of a task fMRI study and data outputs

In a task-based fMRI experiment, each subject performs a sequence of stimulus and control periods while brain activity is recorded through the BOLD signal. The scanner acquires a time series of 3D images, each image discretized into uniformly spaced volume elements (voxels). A single study typically contains on the order of 10^5 voxels per brain volume and hundreds to thousands of repeated acquisitions over time (Lindquist, 2008). The resulting data are thus high-dimensional and noisy, with structured temporal and spatial dependencies.

Before statistical modeling, fMRI data is commonly preprocessed to reduce artifacts and make data comparable over time and over subjects. Standard steps include slice-timing correction, motion correction (realignment), and alignment of functional images to a structural image followed by normalization to a template

space. Spatial smoothing is often applied to increase signal-to-noise ratio and to better match modeling assumptions (Lindquist, 2008). These steps are essential for validity, but they also influence the spatial precision of the final inference, especially in group analyses. The typical output of a first-level (single-subject) analysis is a statistical map, such as a t- or z-map, summarizing evidence for a task effect at each voxel. These maps are the basic objects used in later group-level inference and are also the type of derivative data hosted by repositories such as NeuroVault, which is designed to store and share unthresholded statistical maps for re-use and methodological validation (Gorgolewski et al., 2015).

2.2 From single-subject to group analysis

First-level modeling (within subject)

Consider an experiment comprising N subjects, each scanned over T time points. The brain volume is discretized into voxels, and statistical analysis is restricted to an analysis mask, a subset of voxels used for inference. This mask may be defined a priori (e.g., based on anatomical or functional criteria) or inferred from the data itself, for example by excluding non-brain regions or voxels with insufficient signal. Let m denote the number of voxels within the mask.

A common approach to localizing task-related activity is the General Linear Model (GLM). For subject $j \in \{1, \ldots, N\}$ and voxel $i \in \{1, \ldots, m\}$, let $y_{ij} \in \mathbb{R}^T$ denote the observed BOLD time series over T time points. Let $X_j \in \mathbb{R}^{T \times p}$ be the design matrix encoding the experimental paradigm (e.g., stimulus onsets convolved with a hemodynamic response function, plus nuisance regressors such as motion parameters and drift terms). The voxel-wise model is

$$y_{ij} = X_j \beta_{ij} + \varepsilon_{ij}, \tag{2.1}$$

where $\beta_{ij} \in \mathbb{R}^p$ are regression coefficients and ε_{ij} is a noise term capturing measurement error and residual nuisance effects, often modeled with temporal autocorrelation (e.g., an autoregressive model of order 2) (Lindquist, 2008).

For inference, a contrast vector $c \in \mathbb{R}^p$ defines the task effect of interest as the

linear combination

$$\theta_{ij} = c^\top \beta_{ij}, \tag{2.2}$$

and voxel-wise testing is typically formulated as

$$H_{ij} : \ \theta_{ij} = 0 \quad \text{vs.} \quad K_{ij} : \ \theta_{ij} \neq 0. \tag{2.3}$$

The alternative hypothesis K_{ij} can also be specified as one-sided, depending on whether the interest lies in detecting increases or decreases in activation relative to baseline. Intuitively, c selects and weights the relevant regressors in the design matrix. For example, to test whether a stimulus condition differs from baseline, one can use a contrast that assigns weight 1 to the stimulus regressor and weight 0 to baseline (or implicit baseline) terms, so that θ_{ij} represents stimulus-related activation above baseline. As another example, to compare two conditions directly (e.g., condition A versus condition B), one can use a contrast with weights $+1$ and -1 on the corresponding regressors, so that θ_{ij} represents the differential activation between the two conditions.

In a standard first-level analysis, inference on θ_{ij} is commonly based on a voxel-wise t-statistic and its corresponding p-value. This typically involves transforming (pre-whitening) the time series so that the residual errors ε_{ij} are approximately uncorrelated before applying a t-test (Lindquist, 2008). In practice, many public repositories (including NeuroVault) provide derived first-level outputs rather than the raw time series, typically in the form of voxel-wise estimates of the contrast effect θ_{ij} and/or associated test statistics (e.g., the t-statistics t_{ij}).

Second-level modeling (across subjects)

Group analysis aims at population-level inference by combining subject-specific effects. Consider the contrast effects θ_{ij}, defined for each voxel i and subject j, as in Eq. (2.2). A common second-level (random-effects) model assumes

$$\theta_{ij} = \mu_i + \xi_{ij},$$

where μ_i is the population mean effect at voxel i and the random effects ξ_{ij} capture between-subject variability. Typically, the ξ_{ij} are assumed independent across

subjects with mean zero and (approximately) normal distribution. For each voxel i, we then test whether the mean effect is zero, e.g., against a two-sided alternative:

$$H_i^{\text{group}} : \ \mu_i = 0 \quad \text{vs.} \quad K_i^{\text{group}} : \ \mu_i \neq 0. \tag{2.4}$$

These hypotheses are commonly tested using a one-sample, two-sided t-test across subjects at each voxel. This produces a group-level statistical parametric map with voxel-wise t-statistics t_i^{group} and corresponding map of p-values p_i^{group}.

2.3 Why localization is hard in group studies

Localization in group studies is challenging because the mapping from neuronal activity to observed BOLD signal is indirect and noisy, and because the relationship between anatomy and function varies across individuals. In particular, population inference requires registering brains of different sizes and shapes to a common template. Normalization introduces spatial imprecision and blurring, which can lead to mislocalization or loss of signal in small structures (Lindquist, 2008). Smoothing can further spread signal across neighboring voxels, improving detection at the cost of spatial precision.

In addition to anatomical alignment issues, fMRI time series are affected by multiple nuisance components, including scanner drift, physiological noise (e.g., respiration and heartbeat), and head motion. Even small motion can contaminate voxel measurements through mixing of signals across space, and severe motion may require excluding a subject (Lindquist, 2008). These factors contribute to between-subject variability and can cause standard group summaries (e.g., mean effects) to obscure heterogeneity or to be influenced by outliers.

2.4 Cluster inference and spatial specificity in fMRI

As discussed in Chapter 1, fMRI group studies involve testing a very large number of spatial locations, so multiplicity control is unavoidable. A widely used solution is cluster-based inference: one first applies a cluster-forming threshold to a voxel-wise statistic map, identifies connected supra-threshold clusters, and assigns

cluster-level significance using random field theory or permutation-based calibration. Cluster inference is popular and widely available in standard neuroimaging software.

A main limitation of cluster-based reporting is spatial specificity. A significant cluster provides evidence that there is signal somewhere in the cluster, but it does not quantify where within the cluster the signal lies or how large the activated subregion is. As a consequence, large clusters can be easy to detect while being difficult to interpret precisely (the spatial specificity paradox). In addition, selecting a cluster based on the data and then performing further voxel-wise inference inside it using the same dataset is a form of post-selection inference problem; without explicit correction, nominal voxel-wise error guarantees do not apply to such "drill-down" follow-up analyses (Rosenblatt et al., 2018).

2.5 All-Resolutions Inference (ARI): theoretical framework

ARI was developed to support valid post-hoc quantification in settings where regions are defined after inspecting the data (Rosenblatt et al., 2018). Methodologically, ARI builds on the exploratory multiple-testing framework of Goeman and Solari (2011), which uses the closed testing principle to construct simultaneous statements about the number (or proportion) of true discoveries in any user-selected subset of hypotheses.

Closed testing is a general multiple testing procedure, originally proposed by Marcus et al. (1976) to control the FWER. Recall the setting of Section 1.2, where $\mathcal{M} = \{H_1, \ldots, H_m\}$ is a set of null hypotheses, $\mathcal{T} \subseteq \mathcal{M}$ is the unknown subset of true null hypotheses, and $\mathcal{S} \subseteq \mathcal{M}$ is a generic subset. An intersection hypothesis is defined as

$$H_{\mathcal{S}} = \bigcap_{H_i \in \mathcal{S}} H_i$$

and is true when all the considered null hypotheses are true, that is, $\mathcal{S} \subseteq \mathcal{T}$. If $\mathcal{S} = \emptyset$, the corresponding null hypothesis is considered true by convention. For any intersection hypothesis, closed testing requires a valid α-level test (local test),

i.e., a test that rejects $H_{\mathcal{S}}$ while controlling the type I error at level α:

$$\mathbb{P}(H_{\mathcal{S}} \text{ is rejected by the local test} \mid H_{\mathcal{S}} \subseteq \mathcal{T}) \leq \alpha.$$

Then closed testing rejects a null hypothesis H_i with FWER control only if the local test rejects $H_{\mathcal{S}}$ for every subset $\mathcal{S}$ containing H_i.

Goeman and Solari (2011) exploited this principle to provide simultaneous lower $(1-\alpha)$-confidence bounds for the number of true discoveries in any user-defined subset of hypotheses, as in Eq. (1.1). These are computed as

$$\underline{\mathrm{TD}}(\mathcal{S}) = |\mathcal{S}| - \max_{\mathcal{U} \subseteq \mathcal{S}} \{|\mathcal{U}| \,:\, H_{\mathcal{U}} \text{ is not rejected by closed testing}\}.$$

The key computational challenge is that the number of intersection hypotheses grows exponentially with m. To address this, a shortcut can be used: if the local tests for each intersection hypothesis satisfy certain properties (e.g., using Simes' test (Simes, 1986)), it is sufficient to test only a subset of intersections, which guarantees that the closed testing procedure remains valid. This makes simultaneous inference computationally feasible even in large-scale problems, such as voxel-wise neuroimaging data. This shortcut is implemented for generic multiple testing in the R package `hommel`.

Rosenblatt et al. (2018) adapted this framework to neuroimaging, for hypotheses as in Eq. (2.4), and introduced ARI as a way to quantify signal at multiple spatial resolutions within a single dataset. In particular, ARI provides simultaneous lower confidence bounds on the amount of signal contained in arbitrary voxel sets, which remains valid even when the sets are chosen in a data-driven way (for example, starting from a cluster and subsequently refining it). The neuroimaging-oriented implementation and workflows are provided in the R package `ARIbrain`.

2.6 ROI detection and quantification under ARI

ARI can be used for both detection and quantification of activation in ROI. One may first define candidate regions using external rules (e.g., anatomical ROIs or clusters from a preliminary thresholding step) and then apply ARI to obtain a

guaranteed lower bound on how much signal is present in each region. This way, ARI directly answers questions of the form: "Within this selected set of voxels, what is the minimum amount of true signal compatible with the data at confidence level $1 - \alpha$?".

The key point is that ARI remains valid under post-hoc selection: regions can be defined, refined, and re-defined after seeing the data, and the ARI guarantees still hold. This property is particularly useful in exploratory brain mapping, where analysts naturally iterate between visualization and region definition (Rosenblatt et al., 2018).

Chapter 3

Methodology: voxel-wise prevalence mapping

Building on Chapter 2, this chapter adapts the exploratory multiple-testing framework of Goeman and Solari (2011) to a subject–voxel formulation in order to quantify voxel-wise subject prevalence. The main output is a 3D image of simultaneous lower confidence bounds on prevalence, which supports exploratory visualization and post-hoc reporting (e.g., u-out-of-N maps) without invalidating inference.

3.1 Prevalence objective and hypothesis formulation

The goal of this thesis is to move beyond a purely voxel-wise group null hypothesis and instead quantify subject prevalence: for each voxel, how many subjects show evidence of activation at that location. To connect ARI-style post-hoc inference with this objective, we reformulate the problem in terms of subject–voxel hypotheses.

For each voxel $i \in \{1, \ldots, m\}$ and subject $j \in \{1, \ldots, N\}$, we consider the first-level null hypothesis H_{ij} that the contrast effect at voxel i is absent in subject j, as in Eq. (2.3). This yields a multiple testing problem with mN hypotheses, which can be viewed as an $m \times N$ array indexed by voxel and subject:

$$\begin{Bmatrix} H_{11} & H_{12} & \cdots & H_{1N} \\ H_{21} & H_{22} & \cdots & H_{2N} \\ \vdots & \vdots & & \vdots \\ H_{m1} & H_{m2} & \cdots & H_{mN} \end{Bmatrix}$$

For each voxel i, we focus on the corresponding row set

$$\mathcal{S}_i = \{H_{i1}, H_{i2}, \ldots, H_{iN}\},$$

which collects the N subject-level hypotheses for that voxel. We then apply the exploratory closed testing framework of Goeman and Solari (2011) to obtain simultaneous lower $(1-\alpha)$-confidence bounds on the number of true discoveries in each set $\mathcal{S}_i$. That is, we compute values $\underline{\mathrm{TD}}(\mathcal{S}_i)$ such that

$$\mathbb{P}(\mathrm{TD}(\mathcal{S}_i) \geq \underline{\mathrm{TD}}(\mathcal{S}_i) \text{ for all } i = 1, \ldots, m) \geq 1 - \alpha.$$

In words, with confidence $1-\alpha$, at least $\underline{\mathrm{TD}}(\mathcal{S}_i)$ out of N subjects show a true effect at voxel i, simultaneously for all voxels.

Since $\mathrm{TD}(\mathcal{S}_i)$ equals the number of subjects for which voxel i truly shows activation, the bounds $\underline{\mathrm{TD}}(\mathcal{S}_i)$ provide a conservative voxel-wise estimate of subject prevalence. The simultaneity of these bounds enables two common exploratory reporting strategies without invalidating error control: (i) visualizing prevalence maps to assess which regions have the highest cross-subject consistency, and (ii) selecting a prevalence threshold $u \in \{1, \ldots, N\}$ and reporting only voxels (or voxel clusters) that are active in at least u out of N subjects.

3.2 From repository maps to voxel-wise p-values

In practice, repositories such as NeuroVault provide voxel-wise test-statistic maps rather than raw time series. We therefore start from subject-level t-statistic maps. Let t_{ij} denote the t-statistic for testing H_{ij} at voxel i in subject j. These statistics arise from the first-level GLM contrast test in Eq. (2.3).

For each subject j, the t-statistics t_{ij} follow a Student's t distribution with

degrees of freedom ν_j that depend on the design matrix, mainly on the number T of time points. This distribution can be used to compute p-values p_{ij} in two ways, as following.

t-based calibration (when degrees of freedom are available). If subject-specific degrees of freedom ν_j are available, we compute two-sided p-values as

$$p_{ij} = 2\left(1 - F_{t,\nu_j}(|t_{ij}|)\right),$$

where F_{t,ν_j} is the Cumulative Distribution Function (CDF) of a t distribution with ν_j degrees of freedom.

Normal approximation (when degrees of freedom are unavailable). If ν_j is not available (as in the NeuroVault application presented in the next chapter), we use the large-ν normal approximation:

$$p_{ij} = 2\left(1 - \Phi(|t_{ij}|)\right),$$

where Φ denotes the standard normal CDF.

3.3 Closed testing and voxel-wise prevalence bounds

The p-values p_{ij} provide the inputs required by the closed testing procedure of Goeman and Solari (2011). Therefore, we apply the method with Simes local tests (Simes, 1986), as implemented in the R package `hommel`, to the full family of mN subject–voxel hypotheses and then query the voxel-wise sets $\mathcal{S}_i$ to obtain $\underline{\mathrm{TD}}(\mathcal{S}_i)$ for all voxels. By construction, these bounds are simultaneous across voxels.

The `hommel` package requires as input a single vector containing all p-values for the full multiple testing problem. Therefore the mN p-values for each voxel and subject

$$\begin{Bmatrix} p_{11} & p_{12} & \cdots & p_{1N} \\ p_{21} & p_{22} & \cdots & p_{2N} \\ \vdots & \vdots & & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mN} \end{Bmatrix}$$

can be represented as a vector

$$(p_{11}, p_{21}, \ldots, p_{m1}, p_{12}, p_{22}, \ldots\ldots, p_{mN}) .$$

Subsequently, for each voxel i it is necessary to specify the set of indices identifying the positions that correspond to that voxel, i.e., to $\mathcal{S}_i$:

$$(i, m+i, 2m+i, 3m+i, \ldots, (N-1)m+i) .$$

With this information, the package computes the desired bound $\underline{\mathrm{TD}}(\mathcal{S}_i)$.

3.4 Practical workflow and outputs

The computational pipeline is:

(1) Masking. Define an analysis mask and restrict all subject-level t-maps to the m voxels within the mask. If no mask is provided by the study (e.g., via NeuroVault or accompanying materials), construct one directly from the t-maps by excluding voxels that are zero for all subjects and, if necessary, voxels that exhibit potential numerical artifacts such as infinite values in at least one subject.

(2) p-value maps. Convert each subject's t-map to a p-value map using the formulas in Section 3.2.

(3) Arrange hypotheses by voxel. Form an $m \times N$ matrix of masked p-values, where each row corresponds to a voxel, and each column to a subject. Then flatten the mN p-values to match the input format expected by the implementation, as in Section 3.3.

(4) Closed testing and prevalence bounds. Apply `hommel` to obtain $\underline{\mathrm{TD}}(\mathcal{S}_i)$ for all voxels $i \in \{1, \ldots, m\}$, resulting in a vector of length m containing the voxel-wise prevalence bounds. This vector can then be mapped back onto the spatial grid defined by the analysis mask to reconstruct a 3D prevalence-bound image.

(5) u-out-of-N summaries (optional). For any chosen threshold $u \in \{1, \ldots, N\}$, define the thresholded set of voxels activated in at least u subjects,

$$V_u = \{\, i : \underline{\mathrm{TD}}(\mathcal{S}_i) \geq u \,\}, \tag{3.1}$$

and visualize either the continuous bound map $i \mapsto \underline{\mathrm{TD}}(\mathcal{S}_i)$ or the thresholded map V_u. Connected components of V_u can be reported as a descriptive spatial summary. Inferential guarantees remain voxel-wise unless we additionally query $\underline{\mathrm{TD}}(A)$ for cluster-defined voxel sets A.

3.5 Computational implementation and reproducibility

The proposed methodology was implemented in R. In the following chapter, we illustrate its application using subject-level statistical maps obtained from NeuroVault. These maps were downloaded via the NeuroVault application programming interface, which enables programmatic access to data and metadata, and were handled in Neuroimaging Informatics Technology Initiative (NIfTI) format, the standard file format for storing neuroimaging volumes together with their spatial metadata. The resulting prevalence-bound image was written back to NIfTI for visualization and threshold-based summaries.

For reproducibility, we report: the NeuroVault collection identifier and contrast selection criteria; the numbers N of subjects and m of voxels; the mask definition; the p-value transformation; the confidence level α; the prevalence thresholds u used for summaries; and the R packages used for the analysis.

Interactive visualization tool. We developed an interactive Shiny dashboard for exploring the prevalence bounds, including slice navigation, thresholding by minimum prevalence level, and crosshair inspection of $\underline{\mathrm{TD}}(\mathcal{S}_i)$. The application is hosted online[1].

[1] `https://dilabulanik.shinyapps.io/neurovault_2447/`

Chapter 4

Application: NeuroVault noun/verb judgement dataset

This chapter applies the proposed voxel-wise prevalence mapping framework to a publicly available set of subject-level statistical maps. We describe the dataset, masking and p-value construction, present voxel-wise prevalence bounds, and summarize how the inferred spatial extent changes as the prevalence threshold u increases.

4.1 Dataset and contrast selection

We used statistical maps from the NeuroVault repository (Gorgolewski et al., 2015). Specifically, we analyzed NeuroVault collection #2447[1], which is associated with the precision functional mapping dataset of Gordon et al. (2017). The collection comprises ten participants scanned repeatedly and includes task-fMRI maps from three paradigms (motor, implicit memory encoding, and a mixed task alternating visual perception and semantic noun/verb judgments) across a wide range of contrasts. NeuroVault provides both single-session first-level maps and single-subject (second-level) maps; in this collection, first-level estimates correspond to single sessions, whereas second-level estimates correspond to single subjects.

We selected images whose contrast definition was *noun/verb judgement* > *base-*

[1] `https://identifiers.org/neurovault.collection:2447`

line and restricted to maps with single-subject analysis level, yielding $N = 10$ subject-level t-statistic maps for the analysis pipeline described in Chapter 3. All selected images are in a common template space; in our downloaded NIfTI files, the image grid size (from the NIfTI header) was $(n_x, n_y, n_z) = (48, 64, 48)$, for a total of 147,456 voxels.

4.2 Preprocessing: mask and p-value maps

Mask definition and voxel set size

Because no mask was provided with the collection, we constructed an analysis mask as the union of non-zero voxels across the selected subject maps. This choice is intentionally inclusive: it avoids discarding voxels that may contain valid signal for a subset of subjects, which is important when studying inter-subject heterogeneity. At the same time, a union mask can include voxels with inconsistent coverage across subjects (e.g., voxels that are non-zero for only a few participants), which increases the effective search space and may reduce the ability to certify non-zero prevalence at those locations.

Furthermore, a small number of voxels exhibited non-finite values in some subjects, likely arising from numerical instabilities in the estimation procedure (e.g., vanishing variance estimates in the denominator of the test statistic). To ensure well-defined inference, we excluded all voxels that displayed such irregular values in at least one subject.

The resulting mask contained $m = 67{,}224$ voxels, and excluded the remaining 80,232. Figures 4.1 and 4.2 show the t-statistics and the mask for two participants (subjects 1 and 2). Extreme values of the t-statistic indicate stronger evidence against the null hypothesis and thus suggest activation in the corresponding voxel for that subject. Clear between-subject variability can be observed. While the same contrast is considered, the spatial extent and intensity of patterns differ across subjects, highlighting the motivation for prevalence-based inference at the group level.

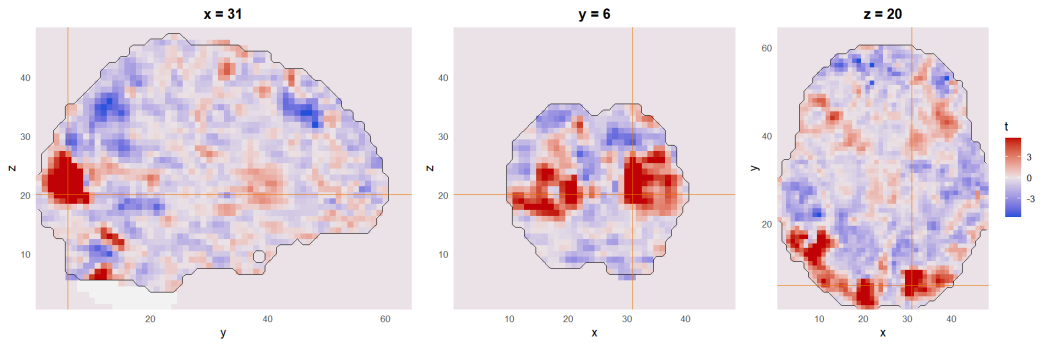

Figure 4.1: Subject 1 t-statistic map for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447).

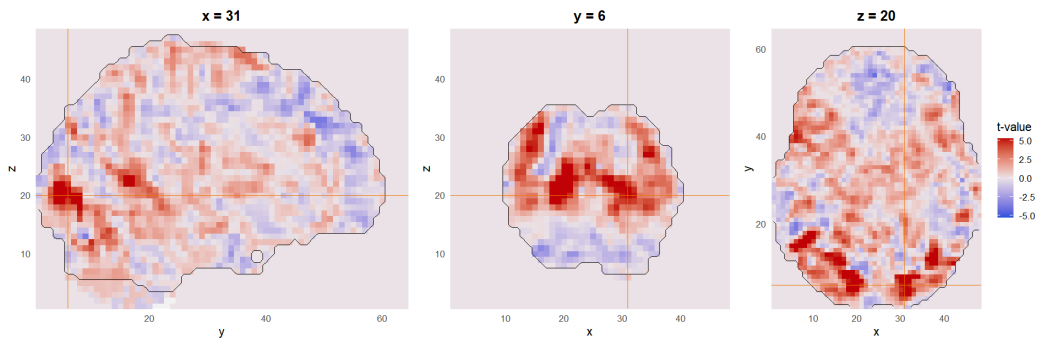

Figure 4.2: Subject 2 t-statistic map for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447).

Construction of subject-level p-values

For each subject, we started from the NeuroVault-provided voxel-wise t-statistic map for the selected contrast. Because the repository provides t-statistic maps but not the corresponding degrees of freedom, we computed approximate two-sided

p-values using the large-ν normal approximation (Section 3.2). This conversion yields a common scale across subjects and provides the p-value inputs required by the closed testing procedure.

The p-value maps for subjects 1 and 2 are shown in Figures 4.3 and 4.4. Smaller p-values indicate stronger evidence against the null hypothesis for the respective voxel and subject.

Subsequently, the resulting p-value maps were restricted to the analysis mask and stacked into an $m \times N$ matrix, with masked voxels indexing rows and subjects indexing columns.

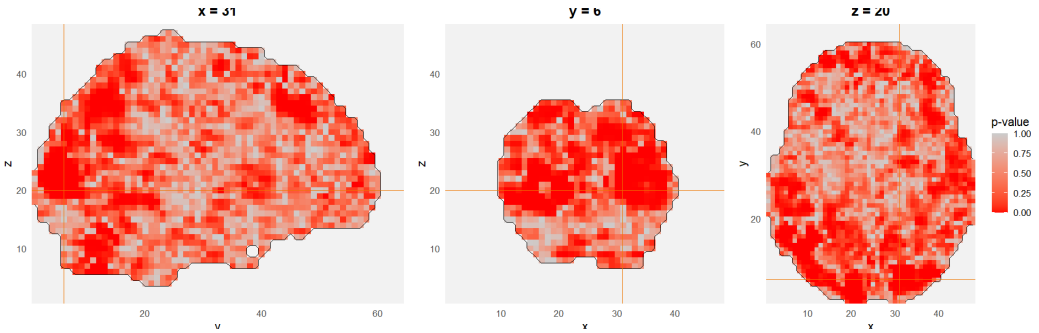

Figure 4.3: Subject 1 p-value map for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447).

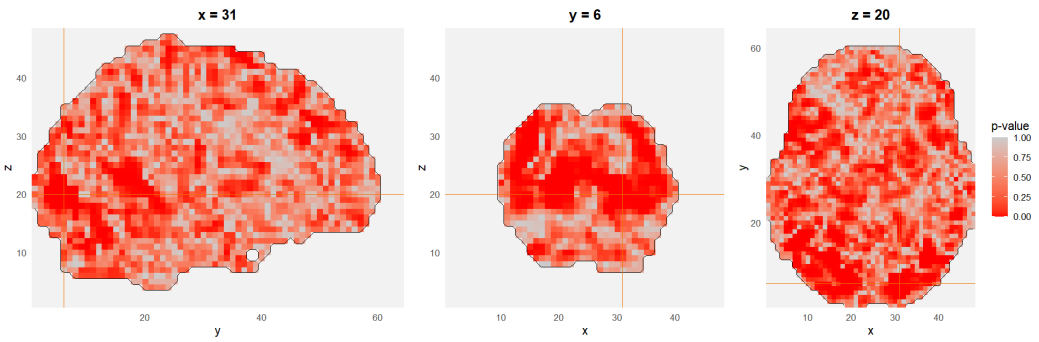

Figure 4.4: Subject 2 p-value map for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447).

4.3 Voxel-wise prevalence bounds

Using the methodology of Chapter 3, we computed voxel-wise lower bounds $\underline{\mathrm{TD}}(\mathcal{S}_i)$ at confidence level $1-\alpha$ with $\alpha = 0.05$ for all voxels in the analysis mask. The output is a 3D image in which each voxel intensity represents the minimum number of subjects that can be claimed active at that location, simultaneously over all voxels (Goeman and Solari, 2011).

Results are shown in Figure 4.5. We summarize the resulting distribution of bounds across voxels and report the corresponding voxel counts and percentages in Section 4.5.

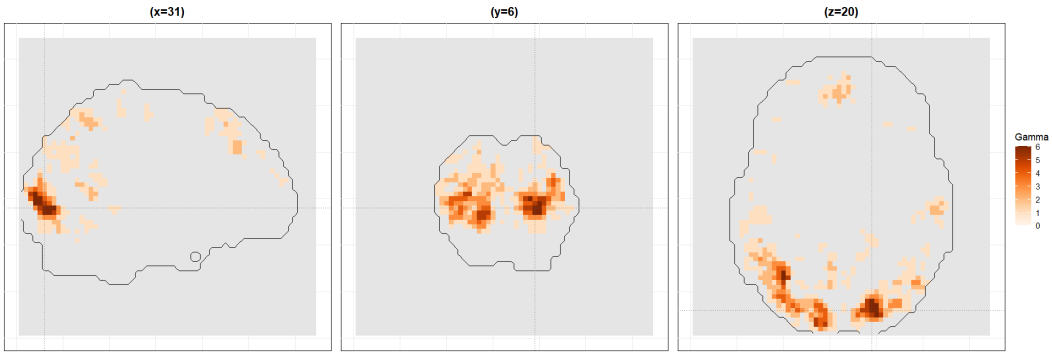

Figure 4.5: Voxel-wise prevalence lower bounds $\underline{\mathrm{TD}}(\mathcal{S}_i)$ for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447), $N = 10$, $\alpha = 0.05$.

4.4 u-out-of-N maps and spatial clustering

To illustrate the effect of prevalence thresholding, we constructed u-out-of-N maps for several values of u. For a chosen threshold u, the thresholded set V_u is the collection of voxels activated in at least u subjects, defined as in Eq. (3.1). As a descriptive spatial summary, we can report clusters, i.e., connected components of V_u, by forming a binary cluster mask (1 inside any connected component, 0

outside) and visualizing orthogonal slices. This highlights spatially contiguous regions where activation is certified in at least u out of N subjects under the simultaneous guarantee.

Figure 4.6 shows clusters for V_5, corresponding to regions of contiguous voxels with activation in at least half of the subjects ($u = 5$ out of $N = 10$). Increasing u would emphasize spatial regions where activation is consistently present across subjects, while reducing u would reveal a larger spatial extent that may be driven by smaller subgroups.

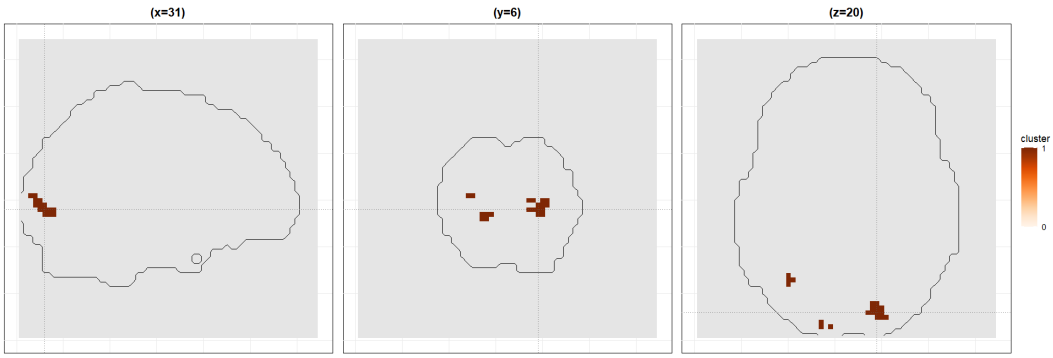

Figure 4.6: Binary cluster mask of the majority-prevalence set $V_5 = \{i : \underline{\mathrm{TD}}(\mathcal{S}_i) \geq 5\}$ for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447), $N = 10$, $\alpha = 0.05$. Voxels inside any connected component are coded as 1 (red) and voxels outside as 0 (background).

4.5 Summary table across prevalence thresholds

To summarize the prevalence pattern, we computed, for each $u \in \{0, \ldots, N\}$, the number of voxels with $\underline{\mathrm{TD}}(\mathcal{S}_i) \geq u$. This yields a concise summary of how much brain volume is supported at each subject-consistency level under a global confidence guarantee. In this dataset, the cumulative volume supported by prevalence bounds decreases rapidly as the threshold u increases. Table 4.1 reports, for each

u, the number of voxels satisfying $\underline{\mathrm{TD}}(\mathcal{S}_i) \geq u$ and the corresponding percentage of the analysis mask.

Table 4.1: Cumulative prevalence summary: number of voxels satisfying $\underline{\mathrm{TD}}(\mathcal{S}_i) \geq u$ and percentage of the analysis mask ($m = 67{,}224$).

u	# voxels with $\underline{\mathrm{TD}}(\mathcal{S}_i) \geq u$	% of mask
1	13,102	19.49
2	4,246	6.32
3	1,477	2.20
4	535	0.80
5	180	0.27
6	60	0.09
7	17	0.03
8	4	0.01
9	0	0.00
10	0	0.00

Approximately 19.5% of voxels in the mask are active in at least one subject. Requiring activity in at least two subjects (a minimal criterion for replicability) reduces this proportion to 6.3%. The number of voxels decreases rapidly as the prevalence threshold u increases: only 0.3% of voxels are active in at least half of the sample, and fewer than 0.1% remain at thresholds from six to eight. No voxels exhibit a lower bound of nine or more active subjects. Overall, the table indicates that while single-subject activations are relatively widespread, strong prevalence across a large fraction of subjects is rare.

4.6 Comparison with ARI

For visual comparison, we additionally applied the ARI method of Rosenblatt et al. (2018), although the two approaches address different inferential targets. Specifically, we considered the group-level null hypotheses of (2.4) and merged information from all subjects to compute group-level p-values through two-sided

one-sample t-tests. Then we identified clusters using a cluster-forming threshold of $\tau = 3.2$, a commonly used value in neuroimaging applications, corresponding approximately to a voxel-wise significance level of 0.001 under a standard normal reference distribution. Out of these, we selected only the clusters with a minimum size of 20 voxels. Finally, we estimated the TDP within each selected cluster using the `ARIbrain` package. Results are shown in Figure 4.7.

Conceptually, the proposed method operates at the voxel level and quantifies between-subject consistency; results can be visualized directly or summarized by identifying clusters of voxels that are active in at least a specified proportion of subjects. In contrast, ARI operates at the level of clusters (or more generally, user-defined regions) and quantifies the proportion of active voxels within each region. Despite these differences in target and interpretation, the resulting maps (Figures 4.5 and 4.7) qualitatively highlight activation in similar brain regions.

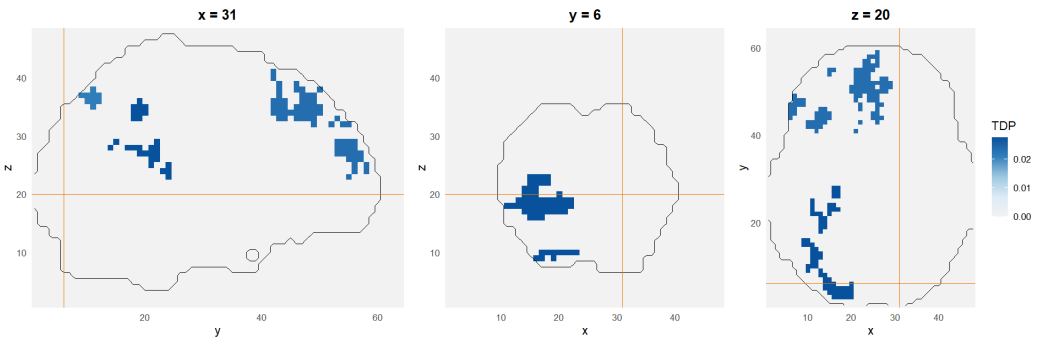

Figure 4.7: ARI cluster-wise lower bounds $\underline{\text{TDP}}(\mathcal{C})$ for the contrast *noun/verb judgement* > *baseline* (NeuroVault collection #2447), $N = 10$, $\alpha = 0.05$. Clusters are identified with a threshold $\tau = 3.2$ and minimum size of 20 voxels.

4.7 Discussion of practical choices

Two practical choices are important for interpreting these results. First, the mask was defined as a union of non-zero voxels across subject maps. This is a conservative choice that avoids excluding potentially relevant voxels, but it can include

regions where only a small subset of subjects have non-zero values. Second, p-values were computed using a normal approximation due to missing degrees of freedom information. This approximation is expected to be reasonable for large degrees of freedom, but it may affect calibration when degrees of freedom are small. These considerations motivate careful reporting of dataset metadata and sensitivity analyses when applying the method to other datasets.

Chapter 5

Conclusion and discussion

5.1 Summary of the thesis

This thesis developed a statistical framework for voxel-wise subject prevalence mapping, with the goal of complementing conventional group-level inference by explicitly quantifying inter-subject consistency. Whereas standard group analyses pool information across participants to estimate a mean (population-level) effect, the proposed approach targets a different inferential question: how many subjects show evidence of activation at a given voxel?

Methodologically, we adopted a u-out-of-N perspective conceptually related to partial conjunction hypotheses (Benjamini and Heller, 2008), and achieved valid inference through a closed-testing construction embedded in an exploratory multiple-testing paradigm with post-hoc guarantees (Goeman and Solari, 2011). The resulting output is a 3D map of voxel-wise lower confidence bounds on prevalence, allowing the researcher to summarize consistency while retaining a global confidence guarantee even after data-driven inspection and selection.

5.2 Key empirical findings from the NeuroVault application

In Chapter 4, we applied the proposed framework to the NeuroVault noun/verb judgement dataset (collection #2447; contrast *noun/verb judgement* > *baseline*). The resulting voxel-wise prevalence lower bounds $\underline{\mathrm{TD}}(\mathcal{S}_i)$ provide simultaneous, spatially resolved guarantees on how many subjects show activation at each voxel.

The main empirical finding is that certified prevalence is highly concentrated: increasing the prevalence threshold rapidly restricts the certified set to a small number of spatially contiguous regions. In other words, the analysis distinguishes effects that are broadly shared across subjects from effects that are supported only in smaller subgroups. Detailed voxel counts and percentages across thresholds are reported in Section 4.5.

5.3 Interpretation and practical implications

Prevalence maps provide an interpretable complement to group aggregation: they shift emphasis from average signal strength to reliability across individuals. This distinction is practically important in exploratory neuroimaging, where investigators often navigate among candidate regions, slices, and sub-clusters. A conventional group effect may reflect heterogeneous subject responses or be driven by a minority of participants, whereas a prevalence bound directly addresses how widespread the signal is under a simultaneous guarantee.

The NeuroVault analysis also illustrates a useful workflow: reporting a small set of u-out-of-N maps (Section 4.4) makes the role of the scientific threshold u transparent. Lower thresholds can reveal broader patterns that may be subgroup-driven, while higher thresholds isolate only the most consistently expressed voxels. The accompanying interactive viewer further supports this interpretability by enabling slice-level inspection and threshold exploration in a reproducible manner.

5.4 Limitations

Several limitations should be considered when interpreting the results and generalizing the approach.

Small sample size. The application used $N = 10$ subjects, which inherently limits the granularity of prevalence statements and makes strict thresholds (e.g., $u \geq 8$) difficult to achieve except in very strongly consistent locations. Larger samples would enable more informative prevalence stratification and more stable spatial conclusions.

Mask construction. The analysis mask was defined as the union of non-zero voxels across subjects (Section 4.2). While conservative in the sense of avoiding exclusion of potentially relevant voxels, this choice can include regions that are only supported by a small subset of participants, thereby increasing the number of voxels where no strong prevalence claim can be made.

p-value calibration. Because subject-specific degrees of freedom were unavailable, test statistics were converted to two-sided p-values using a normal approximation (Section 4.2). This is plausibly reasonable for large degrees of freedom, but it may be miscalibrated for smaller degrees of freedom and could affect the resulting prevalence bounds.

Dependence and spatial specificity. Neuroimaging data exhibit strong spatial dependence and are influenced by preprocessing (registration quality, smoothing, etc.). While the inference targets simultaneous guarantees, the practical spatial appearance of thresholded sets can still be sensitive to these choices, motivating sensitivity analyses and careful reporting.

5.5 Future work

Several extensions would strengthen both methodology and applicability.

Improved subject-level calibration. Whenever possible, extracting (or reconstructing) subject-specific degrees of freedom would allow conversion to p-values using the appropriate reference distribution (t-based rather than normal) and would support more robust calibration checks.

Scaling to larger datasets and pipelines from raw data. An important next step is end-to-end evaluation using raw time-series data to generate first-level maps under standardized preprocessing, followed by prevalence inference. This would clarify how preprocessing choices affect prevalence bounds and would broaden the framework's use in large cohorts.

Covariates and subgroup prevalence. Prevalence inference stratified by covariates (or learned subgroups) would help move from global heterogeneity summaries to explanatory analyses (e.g., identifying which participant characteristics are associated with higher prevalence).

Beyond voxel-wise prevalence. As a natural extension, the framework could be used to investigate consistency across contrasts rather than across subjects, as suggested by Benjamini and Heller (2008). In this setting, the research question becomes: how many contrasts show evidence of activation at a given voxel? After defining the corresponding null hypotheses (grouped across subjects for each contrast), the methodology proceeds unchanged, providing lower confidence bounds on the number of contrasts exhibiting activation at each voxel.

5.6 Concluding remarks

Group-level averaging remains a powerful tool for detecting population-mean effects, but it does not directly quantify how consistently those effects occur across individuals. This thesis introduced a closed-testing-based framework for voxel-wise subject prevalence mapping that provides selection-aware, simultaneous guarantees in exploratory settings (Goeman and Solari, 2011). The NeuroVault application demonstrated that stringent prevalence requirements can dramatically reduce spatial extent, making explicit the difference between broadly shared and subgroup-driven signals. Overall, prevalence-based consistency mapping offers a principled and interpretable complement to standard group inference, supporting more transparent conclusions in the presence of inter-subject variability.

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Finally, to my sister Eda and my mother Kamile: your unconditional love and support have been my greatest motivation throughout this journey, like in every journey I've taken in my life. Thank you for always being by my side; I am truly lucky to have two such strong women behind me. I share this achievement with you. *İyi ki varsınız...*

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Appendix

In this appendix, we repeat the analysis presented in Chapter 4 for the same study but for a different contrast, namely *left hand* > *right hand*. The sample size remains $N = 10$ subjects, and the confidence level is fixed at $\alpha = 0.05$. The analysis mask was recomputed for this contrast following the same procedure described in Chapter 4, resulting in $m = 82{,}984$. All subsequent methodological steps remain unchanged. Below, we report the same set of figures and tables as in Chapter 4 for this contrast.

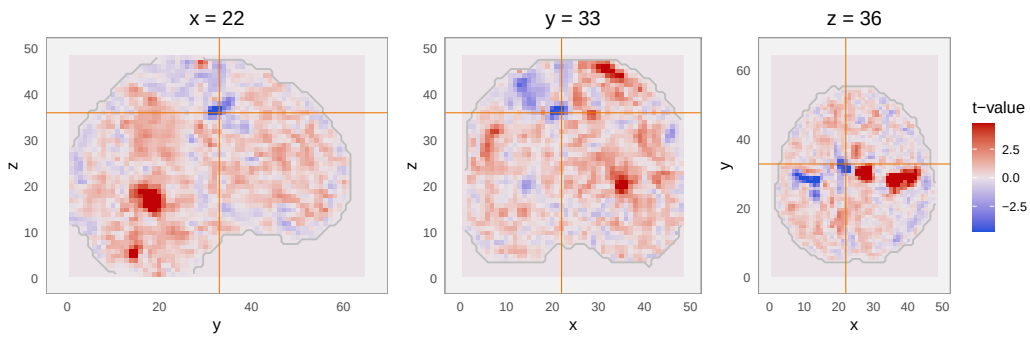

Figure 1: Subject 3 t-statistic map for the contrast *left hand* > *right hand* (NeuroVault collection #2447).

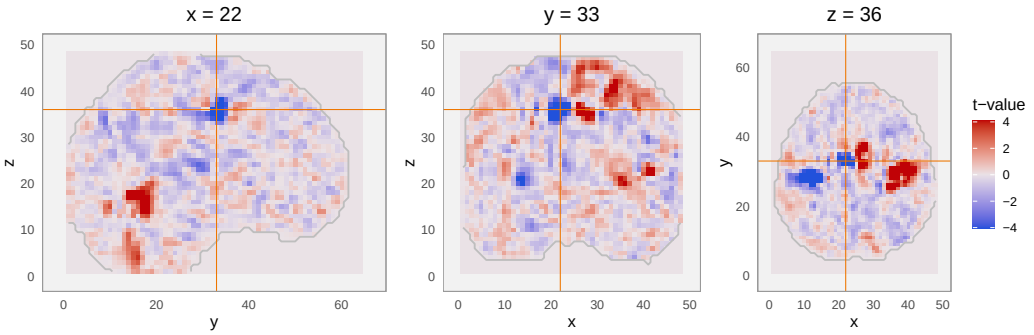

Figure 2: Subject 4 t-statistic map for the contrast *left hand* > *right hand* (NeuroVault collection #2447).

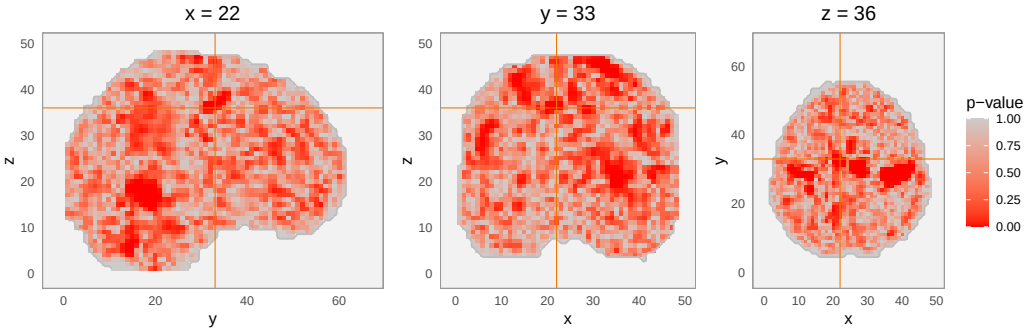

Figure 3: Subject 3 p-value map for the contrast *left hand* > *right hand* (NeuroVault collection #2447).

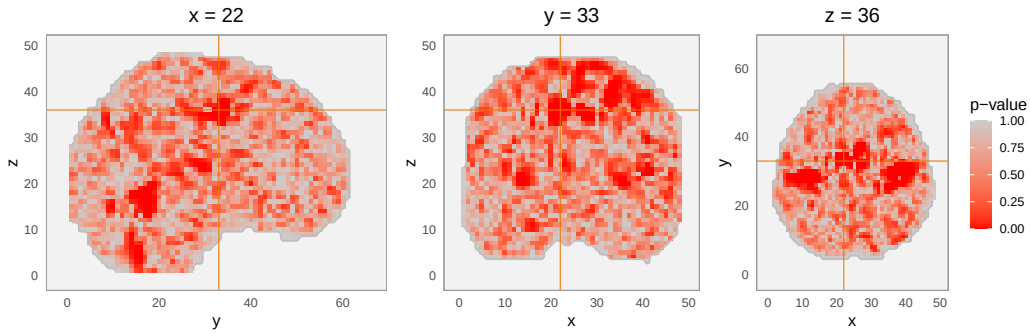

Figure 4: Subject 4 p-value map for the contrast *left hand* > *right hand* (NeuroVault collection #2447).

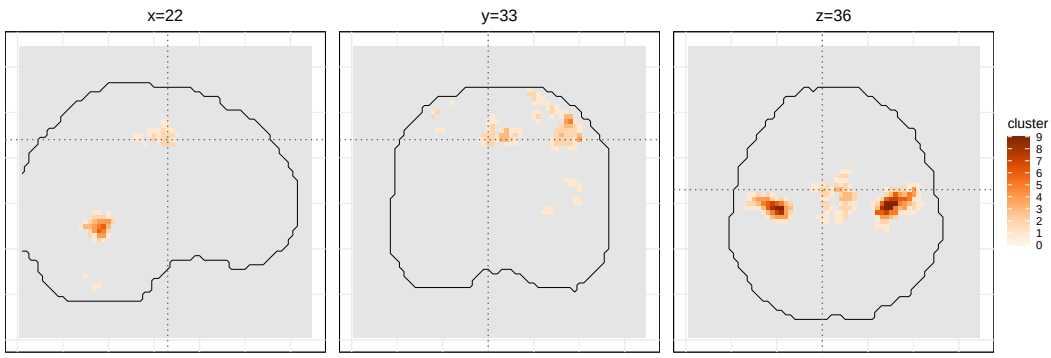

Figure 5: Voxel-wise prevalence lower bounds $\underline{\mathrm{TD}}(\mathcal{S}_i)$ for the contrast *left hand* > *right hand)*, $N = 10$, $\alpha = 0.05$.

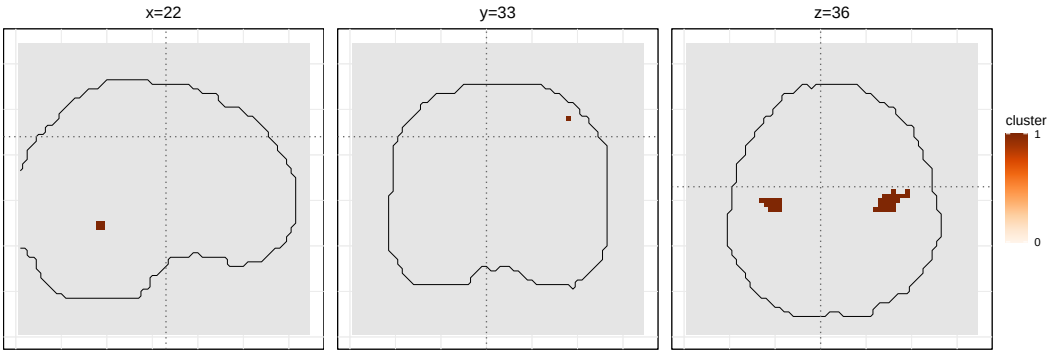

Figure 6: Binary cluster mask of the majority-prevalence set $V_5 = \{\, i : \underline{\mathrm{TD}}(\mathcal{S}_i) \geq 5 \,\}$ for the contrast *left hand* > *right hand* (NeuroVault collection #2447), $N = 10$, $\alpha = 0.05$. Voxels inside any connected component are coded as 1 (red) and voxels outside as 0 (background).

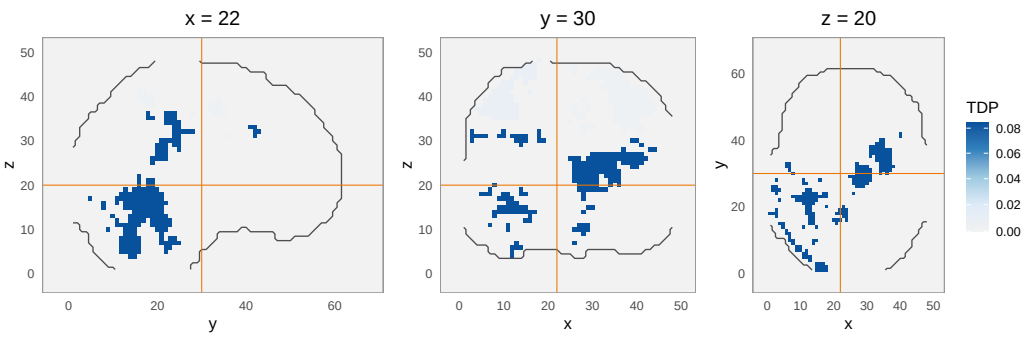

Figure 7: ARI cluster-wise lower bounds $\underline{\mathrm{TDP}}(\mathcal{C})$ for the contrast *left hand* > *right hand* (NeuroVault collection #2447), $N = 10$, $\alpha = 0.05$. Clusters are identified with a threshold $\tau = 3.2$ and minimum size of 20 voxels.

Table 1: Cumulative prevalence summary: number of voxels satisfying $\underline{\mathrm{TD}}(\mathcal{S}_i) \geq u$ and percentage of the analysis mask ($m = 82{,}984$).

u	# voxels with $\underline{\mathrm{TD}}(\mathcal{S}_i) \geq u$	% of mask
1	3,022	3.64
2	1,608	1.94
3	1,029	1.24
4	725	0.87
5	483	0.58
6	318	0.38
7	195	0.23
8	111	0.13
9	36	0.04
10	6	0.01